Repeated Transaction Detection – Python Program

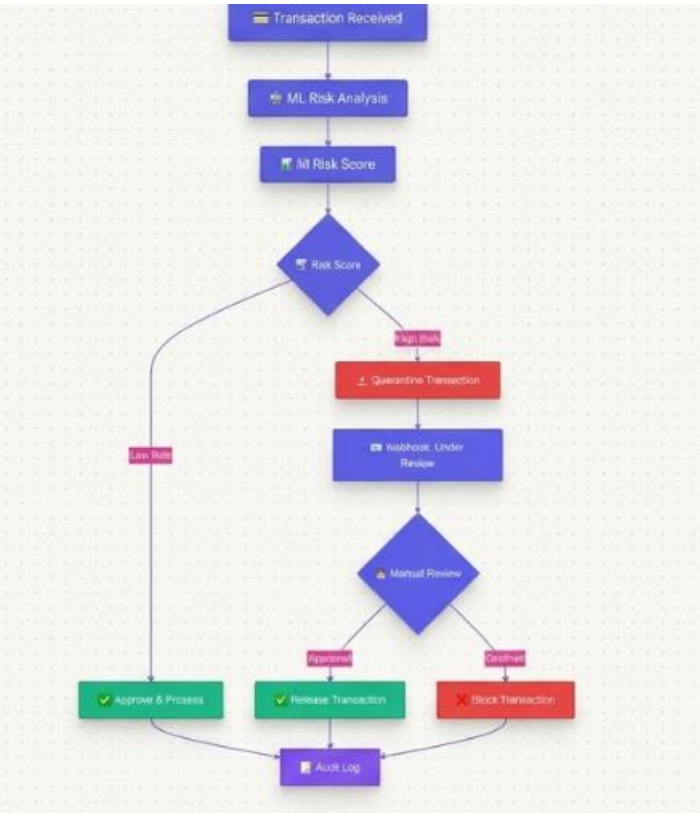
1. Problem Statement

Develop a Python program to process transaction IDs and amounts, detect duplicate transactions, identify transactions exceeding a specified amount limit, and generate a list of suspicious transactions.

2. Algorithm

1. Start the program.
2. Enter the number of transactions.
3. Enter transaction ID and amount.
4. Store all transactions in a list.
5. Check for duplicate transaction IDs.
6. Check whether the amount exceeds the threshold.
7. Add suspicious transactions to a list.
8. Display duplicate and high-value transactions.
9. Stop.

3. Flowchart



4. Python Source Code

transactions = []suspicious = []seen = set()duplicates = []n = int(input("Enter number of transactions: "))limit = float(input("Enter amount threshold: "))for i in range(n): print("\nTransaction", i + 1) tid = input("Enter transaction ID: ") amount = float(input("Enter amount: ")) transactions.append((tid, amount)) if tid in seen: duplicates.append((tid, amount)) suspicious.append((tid, amount, "Duplicate ID")) else: seen.add(tid) if amount > limit: suspicious.append((tid, amount, "High Amount"))print("\n--- TRANSACTION REPORT ---")print("\nDuplicate Transactions:")for t in duplicates: print(t)print("\nSuspicious Transactions:")for t in suspicious: print(t)

5. Sample Input

Enter number of transactions: 4

Enter amount threshold: 5000

Transaction 1

Enter transaction ID: T101

Enter amount: 2000

Transaction 2

Enter transaction ID: T102

Enter amount: 7000

Transaction 3

Enter transaction ID: T101

Enter amount: 3000

Transaction 4

Enter transaction ID: T103

Enter amount: 8000

6. Sample Output

--- TRANSACTION REPORT ---

Duplicate Transactions:

('T101', 3000.0)

Suspicious Transactions:

('T102', 7000.0, 'High Amount')

('T101', 3000.0, 'Duplicate ID')

('T103', 8000.0, 'High Amount')

Total Transactions: 4

Duplicate Count: 1

Suspicious Entries: 3

7. Result

The program successfully detects duplicate transaction IDs, identifies transactions exceeding the specified amount threshold, and generates a suspicious transaction report.